Answer Key & Brief Solutions

1. **Inscribed** angle is half the arc: $\angle ACB = \frac{1}{2} \cdot 86° = 43°$.
2. $\angle APB = \dfrac{144°}{2} = 72°$.
3. Arc measure $= 2 \times 27° = 54°$.
4. Inscribed angles intercepting the same arc are equal $\Rightarrow 41°$.
5. $\angle ACB = \dfrac{1}{2} \cdot 124° = 62°$.
6. Slope $m = \dfrac{-3-5}{2-(-6)} = \dfrac{-8}{8} = -1$.
7. Point–slope: $y - 1 = -\frac{3}{2}(x-4) \Rightarrow y = -\frac{3}{2}x + 6 + 1 = -\frac{3}{2}x + 7$.
8. $3x - 4y = 20 \Rightarrow -4y = 20 - 3x \Rightarrow y = \frac{3}{4}x - 5$; y-intercept is -5.
9. x-int $= -5 \Rightarrow (-5, 0)$; y-int $= 4 \Rightarrow (0, 4)$. Slope $m = \dfrac{4-0}{0-(-5)} = \dfrac{4}{5}$, so $y = \dfrac{4}{5}x + 4$.
10. Slopes $m_1 = 2$ and $m_2 = -\frac{1}{2}$ satisfy $m_1 m_2 = -1 \Rightarrow$ **perpendicular**.
11. $y = 3x - 5$ into $2x + y = 10$: $2x + 3x - 5 = 10 \Rightarrow 5x = 15 \Rightarrow x = 3$, $y = 4$.
12. $y = -x + 8$ into $4x - 2y = 6$: $4x - 2(-x+8) = 6 \Rightarrow 4x + 2x - 16 = 6 \Rightarrow 6x = 22 \Rightarrow x = \frac{11}{3}$, $y = -\frac{11}{3} + 8 = \frac{13}{3}$.
13. $y = 2x + 1$ into $5x - 2y = 11$: $5x - 2(2x+1) = 11 \Rightarrow x - 2 = 11 \Rightarrow x = 13$, $y = 27$.
14. $y = \frac{1}{2}x - 3$ into $3x + 2y = 4$: $3x + 2(\frac{1}{2}x - 3) = 4 \Rightarrow 3x + x - 6 = 4 \Rightarrow 4x = 10 \Rightarrow x = \frac{5}{2}$, $y = \frac{1}{2} \cdot \frac{5}{2} - 3 = \frac{5}{4} - 3 = -\frac{7}{4}$.
15. $y = 4 - 3x$ into $6x + 2y = 1$: $6x + 2(4 - 3x) = 1 \Rightarrow 6x + 8 - 6x = 1 \Rightarrow 8 = 1$; contradiction $\Rightarrow$ **no solution** (inconsistent).
16. Equal sides 10, base 12. Median splits base into 6 and 6. Height $h = \sqrt{10^2 - 6^2} = \sqrt{100 - 36} = 8$. Area $= \frac{1}{2} \cdot 12 \cdot 8 = 48$.
17. $AB = AC = 13$, $BC = 10$. Half-base $= 5$. Height $h = \sqrt{13^2 - 5^2} = \sqrt{169 - 25} = 12$. Area $= \frac{1}{2} \cdot 10 \cdot 12 = 60$.
18. $s = 20$, $b = 16$. Half-base $= 8$. Height $h = \sqrt{20^2 - 8^2} = \sqrt{400 - 64} = \sqrt{336} = 4\sqrt{21}$. Area $= \frac{1}{2} \cdot 16 \cdot 4\sqrt{21} = 32\sqrt{21}$.
19. Area $A = 96 = \frac{1}{2}\, b\, h = \frac{1}{2} \cdot 12 \cdot h \Rightarrow h = 16$. Half-base $= 6$. Equal side $s = \sqrt{h^2 + 6^2} = \sqrt{256 + 36} = \sqrt{292} = 2\sqrt{73}$.
20. Perimeter $2x + 16 = 64 \Rightarrow x = 24$. Half-base $= 8$. Height $h = \sqrt{x^2 - 8^2} = \sqrt{576 - 64} = \sqrt{512} = 16\sqrt{2}$. Area $= \frac{1}{2} \cdot 16 \cdot 16\sqrt{2} = 128\sqrt{2}$.

21. Add equations: $(2x+3y)+(4x-3y)=7+1 \Rightarrow 6x=8 \Rightarrow x=\frac{4}{3}$; then $2(\frac{4}{3})+3y=7 \Rightarrow \frac{8}{3}+3y=7 \Rightarrow 3y=\frac{13}{3} \Rightarrow y=\frac{13}{9}$.
22. Add: $(3x-5y)+(6x+5y)=11+19 \Rightarrow 9x=30 \Rightarrow x=\frac{10}{3}$; then $3(\frac{10}{3})-5y=11 \Rightarrow 10-5y=11 \Rightarrow y=-\frac{1}{5}$.
23. Add: $(7x+2y)+(5x-2y)=5+19 \Rightarrow 12x=24 \Rightarrow x=2$; then $7(2)+2y=5 \Rightarrow 14+2y=5 \Rightarrow y=-\frac{9}{2}$.
24. Multiply first eqn by 3: $12x+18y=24$; add to $-6x+9y=39$ gives $6x+27y=63 \Rightarrow 2x+9y=21$. Also from original $4x+6y=8 \Rightarrow 2x+3y=4$. Subtract: $(2x+9y)-(2x+3y)=21-4 \Rightarrow 6y=17 \Rightarrow y=\frac{17}{6}$; then $2x+3(\frac{17}{6})=4 \Rightarrow 2x+\frac{17}{2}=4 \Rightarrow 2x=-\frac{9}{2} \Rightarrow x=-\frac{9}{4}$.
25. Multiply first eqn by 2: $10x+8y=2$; add to $-10x+6y=44$ gives $14y=46 \Rightarrow y=\frac{23}{7}$. Then $5x+4(\frac{23}{7})=1 \Rightarrow 5x+\frac{92}{7}=1 \Rightarrow 5x=\frac{7-92}{7}=-\frac{85}{7} \Rightarrow x=-\frac{17}{7}$.